Remark 0.1 (Wave-14 reconstitution anchor). This standalone is the inscription of the Platonic holographic-datum heptagon. The closure wave (2026-04-16) added two load-bearing edges to the earlier five-face star. Face (6) is the Drinfeld-centre identification $Z(\mathsf{Rep}_{\mathrm{fact}}(\mathcal{A})) \simeq \mathsf{Rep}_{\mathrm{fact}}(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}))^{\mathsf{E}_2}$ via categorified bar-cobar with half-braiding (cross-ref thm:universal-holography-master, Vol II `chapters/theory/sc_chtop_heptagon.tex`). Face (7) is the derived-algebraic-geometry edge via PTVV 1-shifted symplectic on $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, B\,\mathrm{SC\text{-}Alg})$. Every face is genuinely independent (no tautological routing); the seven classical identifications (operadic, Koszul, factorization, BV–BRST, convolution, Drinfeld-centre, derived-AG) all factor through the chiral Hochschild cochain complex as hub. The topologisation from $\mathrm{SC}^{\mathrm{ch,top}}$ to $\mathsf{E}_3^{\mathrm{top}}$ is PROVED chain-level on the original complex for classes G, L, C, and the critical level (Vol II `chapters/theory/sc_chtop_heptagon.tex`, thm:topologization-tower); for class M it is weight-completed (correct scope, direct-sum chain-level genuinely fails).

THE HOLOGRAPHIC MODULAR KOSZUL DATUM

RAEEZ LORGAT

ABSTRACT. Every 3d holomorphic-topological gauge theory T on $\mathbb{R}_+ \times X$ is controlled by a six-component package $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, \mathcal{C}, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$: boundary chiral algebra, Koszul dual, line-operator category, spectral r-matrix, universal Maurer–Cartan element, and modular shadow connection. We define each component from the bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and its single structural identity $D^2_{\mathcal{A}} = 0$, and compute the datum in full for three families: Heisenberg $\mathcal{H}_k$ (abelian Chern–Simons, class **G**), affine $\widehat{\mathfrak{sl}}_{2,k}$ (non-abelian Chern–Simons, class **L**), and Virasoro Vir_c (3d gravity, class **M**). The bulk–boundary–line triangle is recovered from the datum: the bulk is the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, the boundary is $\mathcal{A}$ itself, and the line category is $\mathcal{A}^!$-mod. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure emerges on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ of derived centre and boundary algebra, not on the bar complex. The central organising principle is the $\mathrm{SC}^{\mathrm{ch,top}}$ *heptagon*: seven edges (operadic / Koszul / factorization / BV–BRST / convolution / Drinfeld centre / derived-AG) identify the datum across presentations, with six classical faces and a seventh derived-algebraic-geometry face inscribed via PTVV on $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, B\,\mathrm{SC\text{-}Alg})$. We construct the anomaly completion: the transgression algebra B_Θ with secondary anomaly $u = \eta^2 = \kappa(\mathcal{A}^!) \cdot \omega_g$, whose vanishing at the critical central charge is the Clifford/exterior bifurcation governing the genus tower.

CONTENTS

1. Introduction

The problem is to organise the algebraic data of a 3d holomorphic-topological (HT) quantum field theory into a single invariant from which every physical observable reconstructs. A 3d HT theory on the half-space $\mathbb{R}_+ \times X$, where X is a smooth algebraic curve, produces three observation algebras: a boundary chiral algebra $\mathcal{A}$ at $\{0\} \times X$, a bulk algebra in the interior, and a category of topological line operators along $\mathbb{R}_+$. The algebraic engine relating these three is Koszul duality in the homotopical modular chiral realm on algebraic curves: the bar construction on $\mathcal{A}$ produces a coalgebra from which the Koszul dual, the r-matrix, and the genus tower are extracted. The governing identity is $D^2_{\mathcal{A}} = 0$; the governing equation is the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

The holographic modular Koszul datum is

$$\mathcal{H}(T) \;=\; (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}).$$

One six-component package, extracted from a single Maurer–Cartan element by a sequence of projections. The components are:

(i) the boundary chiral algebra $\mathcal{A}$;
(ii) the chiral Koszul dual $\mathcal{A}^! = (H^*(\bar{B}(\mathcal{A})))^\vee$;
(iii) the line-operator category $\mathcal{C} \simeq \mathcal{A}^!$-mod;
(iv) the classical r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$;
(v) the universal Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$;
(vi) the modular shadow connection $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$.

The bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an E_1-chiral coassociative coalgebra: it carries a differential (from the chiral product) and a deconcatenation coproduct (from the tensor coalgebra structure). The $\mathrm{SC}^{\mathrm{ch,top}}$ structure does not live on $\bar{B}(\mathcal{A})$; it emerges on the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, where the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is the SC datum with bulk acting on boundary.

Three families illustrate the full range of the datum.

Heisenberg $\mathcal{H}_k$. Abelian Chern–Simons at level k. Shadow class **G**, shadow depth 2. The r-matrix is $r(z) = k/z$ (vanishing at $k = 0$); the line category is Vect; the shadow connection is trivial. This is the base case: all six components are scalar.

Affine $\widehat{\mathfrak{sl}}_{2k}$. Non-abelian Chern–Simons at level k. Shadow class **L**, shadow depth 3. The r-matrix is Yang's $r(z) = k\,\Omega/z$ (trace-form; vanishing at $k = 0$). The line category is $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$ at $q = e^{i\pi/(k+2)}$, recovered from the KZ monodromy. The shadow connection is the KZ connection. Every component is matrix-valued.

Virasoro Vir_c. 3d gravity. Shadow class **M**, infinite depth. The r-matrix is $r_c(z) = (c/2)/z^3 + 2T/z$ (cubic and simple poles); the quartic contact invariant $Q^{\mathrm{contact}} = 10/[c(5c+22)] \neq 0$. The Koszul dual is Vir_{26-c}, self-dual at $c = 13$. The genus tower is infinite; gravity does not truncate.

The bulk–boundary–line (BBL) triangle is not an ad hoc diagram connecting three independent objects. It is a single object (the bar complex with its E_1-chiral coassociative structure) viewed from three angles. The three edges are: boundary⟶bulk (the bar map followed by the trace to the ambient complex), bulk⟶lines (the complementarity cofibre), and boundary⟶lines (Koszul duality). Commutativity of the triangle is Theorem A.

The anomaly completion adjoins to the Koszul dual $\mathcal{A}^!$ a degree-1 generator η with Clifford (not exterior) relation $\eta^2 = \kappa(\mathcal{A}^!)\cdot\omega_g$. The resulting transgression algebra B_Θ has a secondary anomaly $u = \eta^2 = \kappa(\mathcal{A}^!)\cdot\omega_g$ that controls the genus tower: when $u \neq 0$ the Clifford algebra is a matrix algebra and genera are Morita-equivalent; when $u = 0$ the handle operators become exterior generators and the theory sees genuine surface topology.

Conventions. Grading is cohomological ($|d| = +1$). Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar propagator is $d\log E(z,w)$, weight 1 in both variables regardless of field weight. OPE-mode notation: $a_{(n)}b$. The augmentation ideal is $\bar{\mathcal{A}} = \ker(\varepsilon)$; the bar complex uses $\bar{\mathcal{A}}$, not $\mathcal{A}$.

2. The datum

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X.

Definition 2.1 (The holographic modular Koszul datum)**.** The *holographic modular Koszul datum* of $\mathcal{A}$ is the six-component package

$$\mathcal{H}(\mathcal{A}) \;=\; (\mathcal{A},\ \mathcal{A}^!,\ C,\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}),$$

with components defined in Definitions 2.2–2.8 below.

2.1. Boundary algebra.

Definition 2.2 (Boundary algebra)**.** The *boundary algebra* is the chiral algebra $\mathcal{A}$ on X, restricted from the holomorphic boundary $\{0\} \times X \subset \mathbb{R}_+ \times X$ of the 3d HT system.

The boundary carries the full OPE data and is the input to the bar construction. In the 3d HT picture, $\mathcal{A}$ is the boundary condition at the $t = 0$ wall: the holomorphic direction produces the OPE, and the topological direction produces the ordering.

2.2. Bar complex.

The ordered bar complex is

$$\bar{B}^{\mathrm{ord}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}),$$

where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and s^{-1} denotes desuspension ($|s^{-1}v| = |v| - 1$). This is an E_1-chiral coassociative coalgebra: the differential $d_{\bar{B}}$ extracts OPE residues at collision divisors of the Fulton–MacPherson space $\overline{C}_n(\mathbb{C})$, and the deconcatenation coproduct

$$\Delta(a_1 \otimes \cdots \otimes a_n) \;=\; \sum_{p=0}^{n} (a_1 \otimes \cdots \otimes a_p) \otimes (a_{p+1} \otimes \cdots \otimes a_n)$$

splits an ordered tensor sequence at a cut point.

The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$ is the Σ_n-coinvariant image under the averaging map $\mathrm{av}\colon \bar{B}^{\mathrm{ord}} \to \bar{B}^{\Sigma}$. The descent from ordered to symmetric is R-matrix-twisted: $\bar{B}^{\Sigma}_n = (\bar{B}^{\mathrm{ord}}_n)^{R\text{-}\Sigma_n}$. The ordered bar is the primitive object; the symmetric bar is its coinvariant shadow.

Warning 2.3 (Bar complex is not an SC-coalgebra)**.** The bar complex $\bar{B}(\mathcal{A})$ is an E_1-chiral coassociative coalgebra. It does *not* carry an $\mathrm{SC}^{\mathrm{ch,top}}$ structure. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure emerges on the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, computed using the bar complex as a resolution. The pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is the SC datum.

2.3. Koszul dual.

Definition 2.4 (Chiral Koszul dual)**.** The *chiral Koszul dual* of $\mathcal{A}$ is

$$\mathcal{A}^! \;=\; \big(H^*(\bar{B}(\mathcal{A}))\big)^{\vee},$$

the graded linear dual of the bar cohomology coalgebra $\mathcal{A}^{\text{i}} = H^*(\bar{B}(\mathcal{A}))$.

The passage from $\mathcal{A}$ to $\mathcal{A}^!$ involves three steps: bar construction ($\mathcal{A} \mapsto \bar{B}(\mathcal{A})$, coalgebra), cohomology ($\bar{B}(\mathcal{A}) \mapsto \mathcal{A}^{\text{i}}$, dual coalgebra), and Verdier duality ($\mathcal{A}^{\text{i}} \mapsto \mathcal{A}^! = (\mathcal{A}^{\text{i}})^{\vee}$, dual algebra). These are three distinct functors producing four distinct objects:

$$\mathcal{A} \quad\longmapsto\quad \bar{B}(\mathcal{A}) \quad\longmapsto\quad \mathcal{A}^{\text{i}} = H^*(\bar{B}(\mathcal{A})) \quad\longmapsto\quad \mathcal{A}^! = (\mathcal{A}^{\text{i}})^{\vee}.$$

The cobar construction $\Omega(\bar{B}(\mathcal{A})) \simeq \mathcal{A}$ is *inversion*, not Koszul duality.

2.4. Line-operator category.

Definition 2.5 (Line-operator category)**.** The *line-operator category* is

$$\mathcal{C} \;\simeq\; \mathcal{A}^!\text{-}\mathsf{mod}.$$

Objects are topological defect lines along the $\mathbb{R}_+$-direction; morphisms are junction operators. The braiding on $\mathcal{C}$ comes from crossing in the holomorphic plane: two line operators separated in $\mathbb{C}_z$ can be exchanged by moving one around the other. On the Koszul locus, the r-matrix $r(z)$ determines the braiding explicitly.

2.5. Spectral r-matrix.

Definition 2.6 (Classical r-matrix)**.** The *spectral r-matrix* is the binary genus-0 collision residue

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \;\in\; \mathcal{A}^! \,\widehat{\otimes}\, \mathcal{A}^![[z^{-1}, z]].$$

The r-matrix satisfies the classical Yang–Baxter equation (CYBE), which follows from the Arnold relation on $\overline{M}_{0,4}$: the three boundary divisors of the genus-0 4-pointed moduli space correspond to channels (12|34), (13|24), (14|23), and evaluating the Arnold relation on $\Omega \otimes \Omega \in (\mathcal{A}^!)^{\otimes 3}$ produces the three terms of $\sum [r_{ij}, r_{ik}] = 0$.

The r-matrix lives in the E_1 convolution algebra $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$; the modular characteristic $\kappa(\mathcal{A})$ is its Σ_2-coinvariant projection. For abelian algebras, $\mathrm{av}(r(z)) = \kappa$ directly. For non-abelian affine Kac–Moody algebras, the full modular characteristic includes the Sugawara shift:

$$\kappa(V_k(\mathfrak{g})) \;=\; \mathrm{av}(r(z)) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee},$$

where the $\dim(\mathfrak{g})/2$ term is the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$.

2.6. **Universal Maurer–Cartan element.**

Definition 2.7 (Universal MC element)**.** The *universal Maurer–Cartan element* is

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_0 \;\in\; \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}),$$

where $D_{\mathcal{A}}$ is the full bar differential and d_0 is its linear (genus-0, degree-1) part.

The element $\Theta_{\mathcal{A}}$ is always MC because $D^2_{\mathcal{A}} = 0$:

$$D\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

This is the bar-intrinsic construction: $\Theta_{\mathcal{A}}$ exists for every chirally Koszul algebra, at all degrees, as an inverse-limit $\Theta_{\mathcal{A}} = \varprojlim \Theta^{\le r}_{\mathcal{A}}$. The modular characteristic, the r-matrix, the shadow connection, and the full genus tower are projections of $\Theta_{\mathcal{A}}$.

2.7. **Shadow connection.**

Definition 2.8 (Shadow connection)**.** The *modular shadow connection* on $\overline{\mathcal{M}}_{g,n}$ is

$$\nabla^{\mathrm{hol}}_{g,n} \;=\; d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}}),$$

where $\mathrm{Sh}_{g,n}$ is the shadow projection to the (g,n)-component of the modular convolution algebra.

Flatness of ∇^{hol} is the MC equation projected to (g,n). At genus 0, $\nabla^{\mathrm{hol}}_{0,n}$ is unconditionally flat; at genus $g \ge 1$, the connection acquires curvature $\kappa(\mathcal{A}) \cdot \omega_g$ (the genus-g anomaly).

The shadow connection encodes the full genus tower. Its monodromy representations are the physical amplitudes: at genus 0, the KZ monodromy for affine algebras; the BPZ monodromy for Virasoro.

2.8. **Collision filtration and shadow depth.** The modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ carries a collision filtration $F^0_{\mathrm{coll}} \supset F^1_{\mathrm{coll}} \supset \cdots$ indexed by minimum pairwise distance on $\overline{C}_n(\mathbb{C})$. Depth 0 is free propagation; depth k is k-fold nested collision. The spectral sequence $E_1^{p,*} = H^*(\mathrm{gr}^p_{\mathrm{coll}}\, \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, d_0)$ converges to $H^*(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. Collapse is governed by shadow depth:

Class	Shadow depth	OPE pole order	Representative
G	2	2	Heisenberg $\mathcal{H}_k$
L	3	3	Affine $\widehat{\mathfrak{g}}_k$
C	4	4	$\beta\gamma$ system
M	∞	≥ 4	Virasoro, $\mathcal{W}_N$

The discriminant $\Delta = 8\kappa S_4$ controls truncation: $\Delta = 0$ (classes **G**, **L**, **C**) gives a finite shadow tower; $\Delta \ne 0$ (class **M**) gives an infinite tower. The discriminant is linear in κ, not quadratic.

3. The Heisenberg datum

The Heisenberg algebra $\mathcal{H}_k$ at level k is the abelian current algebra generated by a single field J with OPE

$$J(z)\,J(w) \;\sim\; \frac{k}{(z-w)^2}.$$

This is abelian Chern–Simons for $G = U(1)$ at level k.

Proposition 3.1 (Heisenberg datum). *The holographic datum of $\mathcal{H}_k$ is*

$$\mathcal{H}(\mathcal{H}_k) \;=\; \big(\mathcal{H}_k,\ \mathrm{Sym}^{\mathrm{ch}}(V^*),\ \mathsf{Vect},\ k/z,\ k\cdot\eta\otimes\Lambda,\ d\big).$$

Proof. We verify each component.

(i) Boundary. $\mathcal{A} = \mathcal{H}_k$, the free boson at level k.

(ii) Koszul dual. $\mathcal{A}^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$. This is the curved symmetric chiral algebra with $m_0 = -k\omega$; it is *not* $\mathcal{H}_{-k}$ (the same algebra has the same κ on both sides, but the duals differ as curved objects).

(iii) Lines. $C = \mathsf{Vect}$. The line category is trivial: abelian gauge theory has no nontrivial Wilson lines.

(iv) r-matrix. $r(z) = k/z$. The OPE has a double pole; $d\log$ absorbs one, leaving a simple pole. At $k = 0$: $r = 0$ (abelian limit).

(v) MC element. $\Theta_{\mathcal{H}_k} = k\cdot\eta\otimes\Lambda$, where $\eta = d\log(z_1 - z_2)$ is the Arnold form and Λ is the unique (up to scale) bilinear on V. The MC equation is satisfied because $\mathcal{H}_k$ is class **G**: all operations beyond degree 2 vanish.

(vi) Shadow connection. $\nabla^{\mathrm{hol}} = d$. The shadow connection is flat and trivial: the MC element is scalar, so $\mathrm{Sh}_{g,n}(\Theta_{\mathcal{H}_k})$ is a multiple of the identity for all (g,n). □

Remark 3.2 (Modular characteristic). The modular characteristic is $\kappa(\mathcal{H}_k) = k$. This is the abelian prototype: $\mathrm{av}(r(z)) = \mathrm{av}(k/z) = k = \kappa$. The Koszul complementarity is $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = 0$ (class **G**/**L**/free-field complementarity).

Remark 3.3 (Quantised R-matrix). The quantised R-matrix is $R(z) = \exp(k\ \ /z)$, scalar and automatically satisfying the Yang–Baxter equation. The monodromy $\exp(-2\pi i k)$ is the Aharonov–Bohm phase.

4. The Kac–Moody datum

The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k is generated by currents $J^a(z)$ $(a = 1,\dots,\dim(\mathfrak{g}))$ with OPE

$$J^a(z)\,J^b(w) \;\sim\; \frac{k\,\kappa^{ab}}{(z-w)^2} \;+\; \frac{f^{ab}{}_c\,J^c(w)}{z-w},$$

where κ^{ab} is the Killing form and $f^{ab}{}_c$ are structure constants. This is 3d Chern–Simons for G at level k. We take $\mathfrak{g} = \mathfrak{sl}_2$ for concreteness; the formulae generalise to arbitrary simple $\mathfrak{g}$.

Proposition 4.1 (Affine $\widehat{\mathfrak{sl}}_2$ datum). *The holographic datum of $\widehat{\mathfrak{sl}}_{2k}$ is*

$$\mathcal{H}(\widehat{\mathfrak{sl}}_{2k}) \;=\; \Big(\widehat{\mathfrak{sl}}_{2k},\ \widehat{\mathfrak{sl}}_{2-k-4},\ \mathcal{O}_q^{\mathrm{sh}},\ \frac{k\,\Omega}{z},\ \Theta_k,\ d - \kappa_k\,\omega_g - \cdots\Big),$$

with $q = e^{i\pi/(k+2)}$ and $\kappa_k = \dim(\mathfrak{sl}_2)(k+h^\vee)/(2h^\vee) = 3(k+2)/4$.

Proof. We verify each component.

(i) Boundary. $\mathcal{A} = \widehat{\mathfrak{sl}}_{2,k}$, the affine Kac–Moody vertex algebra at level k. The three generators J^a ($a \in \{e, f, h\}$) produce a triple-pole OPE (double pole from the Killing form, simple pole from the Lie bracket).

(ii) Koszul dual. $\mathcal{A}^! = \widehat{\mathfrak{sl}}_{2,-k-4}$. The Koszul involution acts by $k \mapsto -k - 2h^\vee$ with $h^\vee(\mathfrak{sl}_2) = 2$. The complementarity sum is $\kappa(\widehat{\mathfrak{sl}}_{2,k}) + \kappa(\widehat{\mathfrak{sl}}_{2,-k-4}) = 0$, verified at $k = 0$: $\kappa_0 = 3/2$ and $\kappa_{-4} = -3/2$.

(iii) Lines. $\mathcal{C} = \mathcal{O}_q^{\mathrm{sh}} \simeq \mathrm{Rep}(U_q(\mathfrak{sl}_2))$. The braided tensor category of line operators is the representation category of the quantum group at $q = e^{i\pi/(k+2)}$. The identification is the affine Drinfeld–Kohno theorem: the KZ monodromy representation on conformal blocks agrees with the quantum-group R-matrix action on tensor products of evaluation modules.

(iv) r-matrix. $r(z) = k\,\Omega_{\mathrm{tr}}/z$ in trace-form convention, where Ω_{tr} is the trace-form Casimir. At $k = 0$: $r = 0$ (abelian limit). In Kazhdan–Lusztig (KZ) normalisation: $r(z) = \Omega_{\mathrm{KZ}}/((k + h^\vee)z)$ with $\Omega_{\mathrm{KZ}} = 2h^\vee\,\Omega_{\mathrm{tr}}$ the Killing-normalised Casimir. The two conventions are related by a *level-dependent rescaling of the Casimir*, not by the naive identity $k\,\Omega_{\mathrm{tr}} = \Omega_{\mathrm{KZ}}/(k + h^\vee)$: the latter fails as a rational-function equation in k (at $k = 0$ the left side vanishes, the right side is finite and non-zero). The correct statement is that r_{tr} and r_{KZ} are two presentations of the same underlying operadic datum in different bases, exchanged by $\Omega \mapsto \Omega$ and $k \mapsto k/((k + h^\vee) \cdot 2h^\vee)$. At $k = -h^\vee = -2$: trace-form finite, KZ diverges (Sugawara singularity at critical level).

(v) MC element. $\Theta_k = \Theta_{\widehat{\mathfrak{sl}}_{2,k}}$ is determined by $D_{\mathcal{A}}^2 = 0$. The shadow tower terminates at depth 3 (class **L**): the quartic invariant $S_4 = 0$ (the discriminant $\Delta = 8\kappa S_4 = 0$), so the shadow obstruction tower is finite.

(vi) Shadow connection. At genus 0, degree n, the shadow connection is the KZ connection:

$$\nabla^{\mathrm{hol}}_{0,n} \;=\; d - \sum_{i<j} r_{ij}\, dz_{ij} \;=\; d - \frac{1}{k + h^\vee} \sum_{i<j} \frac{\Omega_{ij}\, dz_{ij}}{z_{ij}},$$

where $dz_{ij} = d(z_i - z_j)$. Flatness of the KZ connection is the MC equation at genus 0, degree n. At genus $g \geq 1$, the connection acquires curvature $\kappa_k \cdot \omega_g$ with $\kappa_k = 3(k+2)/4$. □

Remark 4.2 (Quantum group line category). The identification $\mathcal{C} \simeq \mathrm{Rep}(U_q(\mathfrak{sl}_2))$ is a reconstruction theorem: the monodromy of the genus-0 shadow connection $\nabla^{\mathrm{hol}}_{0,n}$ (the KZ connection) generates a braided tensor category that is equivalent to the representation category of the quantum group. The spectral parameter z in $r(z) = k\,\Omega/z$ is the residual parameter after d log-absorption from the OPE; on evaluation modules, it becomes the additive spectral parameter of the Yangian. The full quantised R-matrix $R(z) = 1 + \quad r(z) + O(\quad^2)$ satisfies the quantum Yang–Baxter equation.

Remark 4.3 (Critical level). At $k = -h^\vee = -2$: $\kappa = 0$, all monodromy is trivial, the degree-2 ordered chiral homology $H^1_{\mathrm{ord},2}$ doubles from 4 to 8 for $\mathfrak{sl}_2$ (finite computation on $V \otimes V$, $\chi_{\mathrm{ord},2} = -4$ conserved), and Koszulness fails for the full symmetric bar, whose cohomology becomes $\Omega^*(\mathrm{Op}_{\mathfrak{sl}_2}(D))$ (infinite-dimensional; Frenkel–Teleman 2006; distinct object from the degree-2 ordered slice). The Feigin–Frenkel centre is the algebra of $\mathfrak{sl}_2$-opers on the formal disk: $Z(V_{-2}(\mathfrak{sl}_2)) \simeq \mathrm{Fun}(\mathrm{Op}_{\mathfrak{sl}_2}(D))$. The shadow connection degenerates, and the line category jumps from semisimple to abelian.

5. The Virasoro datum

The Virasoro algebra Vir_c at central charge c is generated by the stress tensor T with OPE

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} \;+\; \frac{2T(w)}{(z-w)^2} \;+\; \frac{\partial T(w)}{z-w}.$$

This is 3d gravity: $SL(2,\mathbb{R}) \times SL(2,\mathbb{R})$ Chern–Simons at levels $k_\pm = \ell/(4G)$, with $c = 3\ell/(2G)$ (Brown–Henneaux).

Proposition 5.1 (Virasoro datum)**.** *The holographic datum of* Vir_c *is*

$$\mathcal{H}(\mathrm{Vir}_c) \;=\; \big(\mathrm{Vir}_c,\ \mathrm{Vir}_{26-c},\ \mathcal{C}_c,\ r_c(z),\ \Theta_c,\ \nabla^{\mathrm{hol}}_c\big),$$

with

$$\begin{aligned} r_c(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, \\ \kappa_c &= \frac{c}{2}, \\ \nabla^{\mathrm{hol}}_{1,1} &= d \;-\; \kappa_c\,\omega_1 \;-\; \frac{10}{c(5c+22)}\,\delta_4 \;-\; \cdots . \end{aligned} \tag{5.1}$$

Proof. We verify each component.

(i) Boundary. $\mathcal{A} = \mathrm{Vir}_c$, the Virasoro vertex algebra.

(ii) Koszul dual. $\mathcal{A}^! = \mathrm{Vir}_{26-c}$. The Koszul involution is $c \mapsto 26 - c$. The complementarity sum is $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = c/2 + (26-c)/2 = 13$. Self-dual at $c = 13$ (not $c = 26$).

(iii) Lines. $\mathcal{C}_c$ is the line-operator category, determined by the r-matrix monodromy. For the Virasoro algebra, the line category is infinite-depth: the monodromy representation is determined by the full BPZ differential equations, not by a finite-dimensional quantum group.

(iv) r-matrix. The r-matrix $r_c(z) = (c/2)/z^3 + 2T/z$ has a cubic pole (gravitational, from the quartic OPE pole after d log-absorption) and a simple pole (stress-tensor exchange). The quartic contact invariant

$$Q^{\mathrm{contact}} \;=\; \frac{10}{c(5c+22)} \;\neq\; 0$$

is nonzero for all $c \neq 0$, $c \neq -22/5$: this is the obstruction to extending $r(z)$ from degree 2 to degree 4, and its non-vanishing is the signature of class $\mathbf{M}$.

(v) MC element. $\Theta_c = \Theta_{\mathrm{Vir}_c}$ is the infinite-depth MC element. The shadow tower does not truncate: all shadow invariants S_r ($r \geq 2$) are nonzero. The modular characteristic is $\kappa(\mathrm{Vir}_c) = S_2 = c/2$.

(vi) Shadow connection. The shadow connection (5.1) has leading curvature $\kappa_c\,\omega_g = (c/2)\,\omega_g$ at genus g, plus an infinite series of corrections from the shadow obstruction tower. □

Remark 5.2 (The gravitational hierarchy). Shadow class $\mathbf{M}$ means the A_∞ tower $(m_1, m_2, m_3, \ldots)$ is infinite: every higher operation m_k is nonzero. The generating depth is $d_{\mathrm{gen}} = 3$ (the operation m_3 determines all m_k recursively via the MC equation), but the algebraic depth is $d_{\mathrm{alg}} = \infty$ (no operation vanishes). Gravity does not truncate.

Remark 5.3 (Two critical values of c). The two critical central charges play distinct roles:

(i) $c = 13$: Koszul self-duality ($\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$). The complementarity parameter is $\kappa + \kappa' = 13 \neq 0$; self-duality is a symmetry enhancement within the perturbative regime.

(ii) $c = 26$: anomaly cancellation ($\kappa(\mathrm{Vir}_{26-c}^!) = (26-c)/2 = 0$ at $c = 26$). The Clifford/exterior bifurcation (Section 8). This is the bosonic string.

Self-duality and anomaly cancellation are separated by 13 units of central charge.

Remark 5.4 (Primitive coproduct). The transferred coproduct Δ_z^{Vir} is strictly primitive at all degrees: the ghost-number obstruction intrinsic to Drinfeld–Sokolov reduction kills every higher-product-lattice tree correction. All gravitational dynamics resides in the collision residue and the infinite A_∞ product tower; the coproduct channel is identically zero.

6. Comparison table

Component	$\mathcal{H}_k$	$\widehat{\mathfrak{sl}}_{2k}$	Vir_c
$\mathcal{A}$ (boundary)	$\mathcal{H}_k$	$\widehat{\mathfrak{sl}}_{2k}$	Vir_c
$\mathcal{A}^!$ (Koszul dual)	$\mathrm{Sym}^{\mathrm{ch}}(V^*)$	$\widehat{\mathfrak{sl}}_{2-k-4}$	Vir_{26-c}
C (lines)	Vect	$\mathrm{Rep}(U_q(\mathfrak{sl}_2))$	C_c (BPZ)
$r(z)$	k/z	$k\,\Omega/z$	$(c/2)/z^3 + 2T/z$
κ	k	$3(k+2)/4$	$c/2$
$\kappa + \kappa'$	0	0	13
Class	**G**	**L**	**M**
Depth	2	3	∞
Q^{contact}	0	0	$10/[c(5c+22)]$
∇^{hol}	d (trivial)	KZ	BPZ (infinite)
Physics	abelian CS	non-abelian CS	3d gravity

The three families represent three qualitatively distinct regimes. Class **G** is Gaussian: scalar, trivial lines, the datum collapses to a single number k. Class **L** is Lie-theoretic: matrix-valued, finite quantum-group lines, the datum is determined by $\mathfrak{g}$ and k. Class **M** is gravitational: infinite depth, no finite quantum group, the datum requires the full infinite tower Θ_c.

7. The BBL triangle

The bulk–boundary–line (BBL) triangle is the central diagram of 3d holomorphic-topological gauge theory. It is not built from three independent objects connected by ad hoc functors; it is a single object (the bar complex with its E_1-chiral coassociative structure) viewed from three angles.

7.1. The three vertices.

Definition 7.1 (BBL triangle)**.** Let $\mathcal{A}$ be a chirally Koszul vertex algebra. The *bulk–boundary–line triangle* of $\mathcal{A}$ consists of:

(i) *Boundary*: the chiral algebra $\mathcal{A}$ itself, the input to the bar construction.

(ii) *Bulk*: the chiral derived centre

$$\mathcal{A}_{\mathrm{bulk}} \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),$$

computed by chiral Hochschild cochains (not topological Hochschild; the geometry of the curve determines which chiral/topological/categorical Hochschild theory applies).

(iii) *Lines*: the category $C \simeq \mathcal{A}^!\text{-}\mathsf{mod}$ of modules over the chiral Koszul dual.

The $\mathrm{SC}^{\mathrm{ch,top}}$ structure lives on the *pair* $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\mathcal{A})$: the chiral derived centre carries brace operations and a chiral Gerstenhaber bracket, and the boundary algebra $\mathcal{A}$ is acted upon by the bulk via the universal brace. The SC datum is two-coloured with directionality: bulk-to-boundary only, no boundary-to-bulk.

7.2. The three edges.

(i) *Boundary → Bulk* (the bar map). The bar construction maps $\mathcal{A}$ to $\bar{B}(\mathcal{A})$; the trace map $\mathrm{tr}_g\colon \bar{B}^{(g)}(\mathcal{A}) \to \mathbf{C}_g(\mathcal{A})$ projects the bar coalgebra into the ambient complex $\mathbf{C}_g(\mathcal{A}) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}(\mathcal{A}))$. The image $\mathbf{Q}_g(\mathcal{A}) = \mathrm{im}(\mathrm{tr}_g)$ is one half of the complementarity decomposition.

(ii) *Bulk → Lines* (the complementarity cofibre). The complementarity theorem gives

$$\mathbf{C}_g(\mathcal{A}) \;\simeq\; \mathbf{Q}_g(\mathcal{A}) \;\oplus\; \mathbf{Q}_g(\mathcal{A}^!).$$

The cofibre $\mathrm{cofib}(\mathbf{Q}_g(\mathcal{A}) \hookrightarrow \mathbf{C}_g(\mathcal{A})) \simeq \mathbf{Q}_g(\mathcal{A}^!)$ is the dual shadow, controlling the line-operator category through the centre–module adjunction.

(iii) *Boundary → Lines* (Koszul duality). The composite $\mathcal{A} \mapsto \bar{B}(\mathcal{A}) \mapsto H^*(\bar{B}(\mathcal{A})) \mapsto (H^*(\bar{B}(\mathcal{A})))^\vee = \mathcal{A}^!$ is bar-dualize. The module category $\mathcal{A}^!$-mod is the categorical output.

Commutativity of the triangle is Theorem A (the Verdier intertwining at the algebra level).

7.3. Reconstruction from the datum. The datum $\mathcal{H}(\mathcal{A})$ is a complete invariant.

Proposition 7.2 (Recovery theorems)**.** *The following reconstructions hold:*

(R1) Collision residue is twisting. $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) = \tau|_{\bar{B}^2(\mathcal{A})}$ *(the universal twisting morphism restricted to bar degree 2).*

(R2) CYBE from Arnold. *The three boundary divisors of $\overline{M}_{0,4}$ produce the three terms of $\sum [r_{ij}, r_{ik}] = 0$.*

(R3) Affine shadow = KZ. *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$:* $\mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) = (k+h^\vee)^{-1} \sum_{i<j} \Omega_{ij}\, dz_{ij}/z_{ij}$.

(R4) Quartic as L_∞ obstruction. $[\mathfrak{R}^{\mathrm{mod}}_4(\mathcal{A})] = [o_4(\mathcal{A})] \in H^2(\mathcal{A}^{\mathrm{sh}}_{4,0})$ *is the obstruction to extending $r(z)$ from degree 2 to degree 4. Vanishes for* **G**, **L**, **C***; nonzero for* **M**.

Remark 7.3 (BBL for Heisenberg). Boundary = $\mathcal{H}_k$. Bulk = $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k)$, which is 3-dimensional (the naive centre is 1-dimensional; the derived centre sees Ext^1 and Ext^2). Lines = Vect. The triangle degenerates: the bulk-to-lines edge is trivial, and the boundary-to-lines edge is trivial.

Remark 7.4 (BBL for Kac–Moody). Boundary = $\widehat{\mathfrak{sl}}_{2,k}$. Bulk = $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\widehat{\mathfrak{sl}}_{2,k})$, with $\mathrm{ChirHoch}^1(V_k(\mathfrak{sl}_2)) \cong \mathfrak{sl}_2$ and total dimension $\dim(\mathfrak{sl}_2) + 2 = 5$. Lines = $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$. The Drinfeld–Kohno theorem makes the boundary-to-lines edge constructive: the monodromy of the KZ connection generates $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$. The bulk-to-lines edge is the complementarity cofibre: $\mathbf{Q}_g(\widehat{\mathfrak{sl}}_{2,-k-4})$ controls the line-operator anomaly.

Remark 7.5 (BBL for Virasoro). Boundary = Vir_c. Bulk = $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$: chiral Hochschild cohomology concentrated in degrees $\{0, 1, 2\}$ with polynomial Hilbert series (this is amplitude, not virtual dimension). Lines = $\mathcal{C}_c$: the infinite-depth line category, governed by the BPZ equations. The triangle is fully active at all three vertices; the bulk-to-lines edge is controlled by the Virasoro complementarity $\mathbf{Q}_g(\mathrm{Vir}_c) \oplus \mathbf{Q}_g(\mathrm{Vir}_{26-c})$.

8. The anomaly completion

8.1. The transgression algebra. Fix a chirally Koszul algebra $\mathcal{A}$ with Koszul dual $B = \mathcal{A}^!$. The curvature of the bar complex on a genus-g curve is $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$. The Koszul dual carries dual curvature $\lambda = \kappa(B) \cdot \omega_g$.

Definition 8.1 (Transgression algebra)**.** The *transgression algebra* is

$$B_\lambda \;=\; B\langle \eta \rangle \,/\, (\eta^2 - \lambda), \qquad |\eta| = 1, \quad d_{B_\lambda}(\eta) = 0,$$

where η is a degree-1 generator with Clifford (not exterior) relation $\eta^2 = \lambda = \kappa(B) \cdot \omega_g$, and the differential on B is extended by $d_{B_\lambda}(b) = d_B(b) + [\eta, b]$ for $b \in B$.

The relation $\eta^2 = \lambda$ is Clifford when $\lambda \neq 0$ and exterior when $\lambda = 0$. This is the algebraic encoding of the anomaly: $\lambda = 0$ means the theory is anomaly-free, and the algebra structure changes qualitatively.

8.2. The secondary anomaly.

Definition 8.2 (Secondary anomaly). The *secondary anomaly* is

$$u = \eta^2 = \lambda = \kappa(\mathcal{A}^!) \cdot \omega_g.$$

It is closed, central, and acts as the scalar $\kappa(\mathcal{A}^!)$ on standard modules.

The secondary anomaly is the bifurcation parameter for the genus tower. Its behaviour depends on the Koszul complementarity:

Family	$\kappa(\mathcal{A}^!)$	u	Regime
$\mathcal{H}_k$	$-k$	$-k\,\omega_g$	$u = 0$ at $k = 0$
$\widehat{\mathfrak{sl}}_{2k}$	$3(-k-2)/4$	$3(-k-2)\omega_g/4$	$u = 0$ at $k = -2$ (critical)
Vir_c	$(26-c)/2$	$(26-c)\omega_g/2$	$u = 0$ at $c = 26$ (string)

8.3. Genus-g Clifford completion.

Construction 8.3 (Clifford completion). At genus g, adjoin $2g$ handle operators $\alpha_1, \beta_1, \dots, \alpha_g, \beta_g$ with

$$\alpha_i\beta_j + \beta_j\alpha_i = \delta_{ij}\,u, \qquad \alpha_i\alpha_j + \alpha_j\alpha_i = 0, \qquad \beta_i\beta_j + \beta_j\beta_i = 0.$$

The genus-g Clifford completion is the algebra

$$\mathfrak{G}_g(\mathcal{A}) = B_\lambda \otimes \mathrm{Cl}(H_1(\Sigma_g); u).$$

The handle operators record the genus-g topology. Their anticommutator is controlled by the secondary anomaly u, which splits quantum field theory into two qualitative regimes.

8.4. The dichotomy.

Proposition 8.4 (Clifford/exterior dichotomy). *The genus tower of $\mathcal{A}$ bifurcates at $u = 0$:*

(i) When $u \neq 0$: the Clifford algebra $\mathrm{Cl}(H_1(\Sigma_g); u)$ is a matrix algebra (dimension 2^g) after inverting u. Genus-g is Morita equivalent to genus-0. All genus corrections are resummable from genus-0 data. This is the perturbative regime.

(ii) When $u = 0$: the Clifford anticommutator vanishes. The handle operators become exterior generators on $H_1(\Sigma_g)$ (dimension 2^{2g}). The theory sees genuine surface topology. The genus tower collapses ($\kappa_{\mathrm{eff}} = 0$).

Proof. Standard Clifford algebra theory. When the bilinear form $u \cdot \langle\cdot,\cdot\rangle$ on $H_1(\Sigma_g, \mathbb{Z})$ is nondegenerate (i.e., $u \neq 0$), the Clifford algebra is a matrix algebra $M_{2^g}(\mathbb{C})$ (even-dimensional, nondegenerate symplectic). When $u = 0$, the bilinear form degenerates and the Clifford algebra becomes the exterior algebra $\Lambda^*(H_1(\Sigma_g)^*)$. □

8.5. The three families.

Heisenberg. $u = -k\,\omega_g$. At $k = 0$: $u = 0$, exterior regime. At generic k: $u \neq 0$, perturbative. The bifurcation is at the abelian limit.

Kac–Moody. $u = 3(-k-2)\omega_g/4$. At $k = -2$ (critical level): $u = 0$, the Feigin–Frenkel centre appears, the genus tower collapses. At generic k: $u \neq 0$, perturbative. The critical level is the bifurcation point.

Virasoro. $u = (26-c)\omega_g/2$. At $c = 26$: $u = 0$, the bosonic string becomes topological ($Z_{\mathrm{grav}} = 1$). At generic c: $u \neq 0$, perturbative. At $c = 13$: $u = 13\omega_g/2 \neq 0$, so Koszul self-duality lives within the perturbative regime.

The gravitational partition function at genus g is the super-trace of the MC element through the Clifford factor:

$$Z_g \;=\; \mathrm{str}_{\mathfrak{G}_g/B_\Theta}(\Theta_c^{(g)}).$$

The complementarity theorem gives a Lagrangian decomposition

$$H^*(\overline{\mathcal{M}}_g,\, \mathcal{Z}(\mathrm{Vir}_c)) \;\simeq\; \mathbf{Q}_g(\mathrm{Vir}_c) \;\oplus\; \mathbf{Q}_g(\mathrm{Vir}_{26-c})$$

into boundary (+1-eigenspace of Verdier involution) and bulk (−1-eigenspace) sectors. The shifted symplectic structure makes both Lagrangian.

8.6. **The anomaly parameter as holographic observable.** The secondary anomaly u is the single scalar that governs the passage from genus 0 to all genera. For each family, u vanishes at a distinguished value of the level/central charge:

Family	$u = 0$ at	Physical meaning	Regime
$\mathcal{H}_k$	$k = 0$	trivial theory	exterior
$\widehat{\mathfrak{g}}_k$	$k = -h^\vee$	critical level, Feigin–Frenkel	exterior
Vir_c	$c = 26$	bosonic string, anomaly cancellation	exterior

The datum $\mathcal{H}(\mathcal{A})$ together with the anomaly completion B_Θ controls the full 3d HT correspondence: the six components determine the genus-0 physics (the datum), and the secondary anomaly u determines the genus tower (the completion). The entire structure descends from a single identity: $D^2_{\mathcal{A}} = 0$.

9. Concluding remarks

9.1. **The E_n circle.** The holographic datum sits inside the E_n operadic circle:

$$\mathsf{E}_3^{\mathrm{top}}(\text{bulk}) \;\to\; \mathsf{E}_2(\text{boundary chiral}) \;\to\; \mathsf{E}_1(\text{bar/QG}) \;\to\; \mathsf{E}_2(\text{Drinfeld centre}) \;\to\; \mathsf{E}_3^{\mathrm{top}}(\text{derived centre}).$$

Each arrow is: restriction to codimension-2 defect, ordered bar complex, categorified averaging (Drinfeld centre), higher Deligne (derived centre). The circle partly closes: for simple $\mathfrak{g}$, the E_3-topological deformation space of $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ is $H^3(\mathfrak{g})[[\ \ ]]$, one-dimensional; the formal disk restriction recovers the Costello–Francis–Gwilliam Chern–Simons E_3-algebra. The passage from $\mathrm{SC}^{\mathrm{ch,top}}$ to E_3 requires a conformal vector at non-critical level (Sugawara), which makes $\mathbb{C}$-translations Q-exact and topologises the holomorphic direction. Without conformal vector: stuck at $\mathrm{SC}^{\mathrm{ch,top}}$.

9.2. **Scope.** The topologisation $\mathrm{SC}^{\mathrm{ch,top}} \to \mathsf{E}_3^{\mathrm{top}}$ is proved for affine Kac–Moody algebras $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$ (where the Sugawara construction is explicit). For general chiral algebras with conformal vector (Virasoro, $\mathcal{W}$-algebras), the topologisation is cohomological but the chain-level statement is open. The proof is cohomological: for class **M** algebras, where chain-level data is essential, E_3 may exist only on cohomology.

The chiral Hochschild cohomology $\mathrm{ChirHoch}^*(\mathcal{A})$ is concentrated in degrees $\{0, 1, 2\}$ with polynomial Hilbert series (Theorem H). This is cohomological amplitude, not virtual dimension. It is not $\mathbb{C}[\Theta]$, and it is not the Gelfand–Fuchs cohomology (which is infinite-dimensional).

9.3. **From the datum to physics.** The derived chiral centre IS the bulk of a real 3d HT gauge theory. This is the content of the BBL identification: the bar construction on $\mathcal{A}$ produces a coalgebra; the chiral Hochschild cochain complex of $\mathcal{A}$ (computed from the bar resolution) produces the bulk observables; the $\mathrm{SC}^{\mathrm{ch,top}}$ structure on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ encodes the open/closed interaction. The datum $\mathcal{H}(\mathcal{A})$ is the smallest package from which the full physical correspondence reconstructs.

The programme beyond this paper: the global BBL triangle (bulk determines boundary determines bulk, proved boundary-linear; full reconstruction open), the chain-level topologisation for class **M**, and the closing of the E_n circle via the Drinfeld centre identification $Z(\mathbf{U}_{\mathcal{A}}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$.

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